# RAJ GANDHI

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Raj Gandhi | Raj Gandhi (rajgandhi7) | Raj Gandhi | Raj Gandhi

Haifa, 3200003, Israel

## Education

- **Technion - Israel Institute of Technology** *July 2022 – February 2026 (expected)*
  ***Doctor of Philosophy candidate, Department of Mathematics*** Haifa, Israel
  - Ph.D. thesis advisors: Professor Alexander Oron and Professor Alexander Nepomnyashchy
  - Thesis title: Instabilities in nanofluid layers

- **École Centrale de Lyon** *2019 – 2021*
  ***Master (M2 & M1) in Aerospace Propulsion and Space Engineering*** Lyon, France
  - Grade: 13.9/20 (B), Mention: Très Bien/Very good; Top 20%
  - Master thesis: Elastic turbulence for complex fluid – viscoelastic in curvilinear geometries
  - Master thesis grade: 16/20

- **Sardar Vallabhbhai Patel Institute of Technology** *2015 – 2019*
  ***Bachelor of Engineering in Aeronautical Engineering*** India
  - CGPA: 9.01/10

## Research Experience

- **Technion - Israel Institute of Technology** *Haifa, Israel*
  ***Graduate research assistant, MSCA research fellow nanoPaInt*** July 2022 – February 2026 (expected)
  - Developing a mathematical model for thermosolutal instabilities in a nanofluid layer, taking into account the nanoparticle concentration-dependent thermophysical properties
  - Investigating the hydrodynamic stability of a moderately dense nanoparticle suspension
  - Employing the method of asymptotic expansion to investigate the long–wave monotonic and oscillatory thermosolutal instabilities in a moderately dense nanofluid layer
  - Developing a mathematical model for thermosolutal instabilities in a nanofluid layer, taking into account the colloidal particle interaction with the liquid–gas interface and examining the hydrodynamic stability
  - Investigating the role of the thermal conductivity stratification by the direct numerical simulations of a full nonlinear problem
  - Exploiting the nonlinear simulation data and performing FFT, spatio-temporal, and phase plane analysis to investigate different modes of oscillatory thermocapillary instability

- **Technische Universität Darmstadt, Institute of Technical Thermodynamics** *Darmstadt, Germany*
  ***Visiting scholar, MSCA research fellow secondment*** October 2024
  - Nonlinear multiphase model setup and simulations for thermosolutal instabilities in a nanofluid layer in COMSOL Multiphysics software.

- **Laboratory of Fluid Mechanics of Lille** *Lille, France*
  ***Research Intern** – [Master thesis]* April 2021 – September 2021
  - Creating parametric grids and simulations of the Taylor-Couette flow, Serpentine channel flow, and Von Kármán swirling flow geometries using C++ programming in the OpenFOAM.

- Investigating the elastic turbulence in curvilinear geometries using direct numerical simulations in OpenFOAM
- Mathematical post-processing of data, such as Fourier transform, Power-law comparison, and plotting the Temporal power spectra.
- keywords: Elastic turbulence, Elastic instabilities as a function of the Weissenberg numbers, viscoelastic/non-Newtonian fluid, Curvilinear geometries (Taylor-Couette flow and Serpentine channel flow), Kinetic energy spectra comparison with power-law scaling range exponent, and direct numerical simulations in OpenFOAM.

- **Laboratoire de Tribologie et Dynamique des Systèmes** *Lyon, France*
  *Research Intern (Master internship)* May 2020 – July 2020
  - Subject of internship – Fan design for the aeraulic optimization of an air purifier.

## Research Interests

- Theoretical and numerical investigation of the interdisciplinary problems in fluid mechanics
- Hydrodynamic stability in complex fluids: Colloidal dispersion, Viscoelastic, Magnetic Liquids
- Interfacial instabilities, thin film hydrodynamics, asymptotic methods
- Nonlinear stability and pattern-forming instabilities
- Low and high Reynolds number hydrodynamics, effect of stratification in practical applications

## Fellowships

- Technion Departmental Scholarship Portions (July 2025 – December 2025 )
- Marie Skłodowska – Curie Actions, Early stage researcher fellowship (July 2022 – July 2025)
- École Centrale de Lyon partial tuition fees scholarship (2019 – 2021)
- Mukhyamantri Yuva Swavalamban Yojana, 50% tuition fees scholarship (2015 – 2019)

## Publications

J=Journal, S=In Submission, P = In preparation, T=Thesis

**[J.1]** **Gandhi, R.**, Nepomnyashchy, A., & Oron, A. (2025d). Thermosolutal instabilities in a moderately dense nanoparticle suspension. *J. Fluid Mech.*, *1011*, A52. https://doi.org/10.1017/jfm.2025.374

**[J.2]** **Gandhi, R.**, Nepomnyashchy, A., & Oron, A. (2025i). The onset of long–wave oscillatory thermosolutal instability in a heated layer of a moderately dense nanofluid. *Phys. Fluids*, *37*(9), 092120. https://doi.org/10.1063/5.0289945

**[J.3]** **Gandhi, R.**, Nepomnyashchy, A., & Oron, A. (2025a). Longwave thermosolutal instability in a moderately dense nanoparticle suspension. *Eur. Phys. J. Special Topics*, to appear

**[P.1]** **Gandhi, R.**, Nepomnyashchy, A., & Oron, A. (2025e). Thermosolutal instabilities in a moderately dense nanoparticle suspension in the presence of interfacial kinetics: The case of cooling at the substrate [Under consideration for publication in J. Fluid Mech.]

**[P.2]** **Gandhi, R.**, Nepomnyashchy, A., & Oron, A. (2025c). The onset of anomalous solutocapillary instability in a nanofluid layer with interfacial kinetics [Under consideration for publication in J. Fluid Mech.]

**[P.3]** **Gandhi, R.**, Nepomnyashchy, A., & Oron, A. (2025f). Thermosolutal instabilities in a moderately dense nanoparticle suspension in the presence of interfacial kinetics: The case of heating at the substrate [Under consideration for publication in J. Fluid Mech.]

**[P.4]** **Gandhi, R.**, Nepomnyashchy, A., & Oron, A. (2025b). Nonlinear thermocapillary instability in a moderately dense nanofluid layer: Role of the thermal conductivity stratification

**[T.1]** Gandhi, R. (2026, February). *Instabilities in nanofluid layers* [PhD thesis]. Technion - Israel Institute of Technology [In preparation]

**[T.2]** Gandhi, R. (2021). *Elastic turbulence in curvilinear geometries* [Master's thesis]. Laboratoire de Mecanique des Fluides de Lille - Kampé de Fériet and École Centrale de Lyon

## Conference Proceedings

**[C.1]** **Gandhi, R.**, Nepomnyashchy, A., & Oron, A. (2024a). Instabilities in a non-isothermal nanofluid layer in a gravity field [oral presentation]. *Bifurcations and Instabilities in Fluid Dynamics, Edinburgh, Scotland*

**[C.2]** **Gandhi, R.**, Nepomnyashchy, A., & Oron, A. (2024b). Instabilities in a non-isothermal nanofluid layer in a gravity field [oral presentation]. *2024 Abarbanel Workshop on applied and computational mathematics, Tel Aviv, Israel*

**[C.3]** **Gandhi, R.**, Nepomnyashchy, A., & Oron, A. (2025h). Thermosolutal instabilities in a moderately dense nanoparticle suspension: Effect of interfacial nanoparticle kinetics [oral presentation]. *12th Conference of the International Marangoni Association, Madrid, Spain*

**[C.4]** **Oron, A.**, Nepomnyashchy, A., & Gandhi, R. (2025). Thermosolutal instabilities in a moderately dense nanoparticle suspension: Effect of thermal conductivity stratification [oral presentation]. *12th Conference of the International Marangoni Association, Madrid, Spain*

**[C.5]** **Gandhi, R.**, Nepomnyashchy, A., & Oron, A. (2025g). Thermosolutal instabilities in a moderately dense nanoparticle suspension [poster presentation]. *IFMC2025, Haifa, Israel*

## Technical Skills

- **Programming Languages:** python, C++
- **Software:** Wolfram Mathematica, Matlab, COMSOL, pyoomph, OpenFOAM, Dedalus, paraView, GitHub

## References

1. **Professor Alexander Oron**
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   *Relationship: Ph.D. thesis Advisor*

2. **Professor Alexander Nepomnyashchy**
   The Vivian and Sidney Konigsberg Academic Chair, Department of Mathematics,
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